15 Prove algebraically that the sum of the squares of any two consecutive even numbers is always a multiple of 4

Let consecutive even numbers be $2n$ and $2n+2$.

Their squared sum is $(2n)^2 + (2n+2)^2$

$= 4n^2 + 4n^2 + 8n + 4$

$= 4(2n^2 + 2n + 1)$, a multiple of 4.

3 marks total
Represent both evens
Expand both squares
Factor out 4

(Total for Question 15 is 3 marks)